Jenny Mai

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Education

University of Illinois Urbana-Champaign Aug 2022 – May 2026

Bachelor of Science in Data Science & Information Sciences with a Computer Science Minor *3.90/4.00 GPA*

Honors: James Scholar and UIUC iSchool Scholarship Recipient

Exchange Study: University of Melbourne and Nanyang Technological University *2024 - 2025*

Experience

Software Engineer Intern Jan 2025 – Present

Bumpups *Remote*

- Design and optimize prompts using DSPy to enhance fine-tuned model performance and improve user interactions.
- Collaborate with software teams to integrate advanced prompting techniques into scalable solutions.
- Optimize input data quality for fine-tuning and prompt development, achieving 90% of prompt output targets.

Software Engineer Intern Aug 2023 – Present

Illini Media *Champaign, IL*

- Develop & maintain backend systems (Flask, Next.js, PostgreSQL, APIs) to optimize publication workflows by 45%.
- Implement role-based access controls and security protocols to ensure data integrity and platform security.
- Optimize database operations and automate workflows through CI/CD pipelines, improving system efficiency.

Project Manager Intern & Fullstack Developer Intern Jul 2024 – Dec 2024

University of Melbourne C-Lara Lab *Melbourne, AU*

- Led a 5-member team to develop a web application for reinforcement learning experiments on Chain of Thought reasoning, implementing Agile methodologies.
- Designed and developed a scalable backend using Django and PostgreSQL, and integrated APIs with a React.js UI.
- Deployed on Google Cloud with GitHub Actions-Docker CI/CD pipelines, reducing manual deploy by 85%.
- Conducted prompt engineering to enhance AI model performance, enabling real-time decision-making.

Research Assistant - Data Analyst Aug 2023 – Present

UIUC Center for Health Informatics | Supervisor: Ian Brooks (UIUC) *Melbourne, AU*

- Led data cleaning and analysis of complex bilingual Pneumococcal Vaccine data using Python and R.
- Identified and resolved data errors and inconsistencies in 25,000+ patient records, ensuring data integrity.
- Conducted exploratory data analysis (EDA) and statistical analysis to generate insights into vaccine effectiveness.

Data Engineer Intern Jun 2021 – Mar 2022

VietAI Research | Supervisors: Thang Luong, Trieu Trinh (Google DeepMind) | research.vietai.org/mtet *San Francisco, CA*

- Scraped and preprocessed 30,000 medical journals using Selenium, boosting NLP model accuracy by +3.5 BLEU.
- Optimized data pairing with quality controls, enhancing English-Vietnamese translation performance.

Projects

Melbourne Real Estate & Prediction | *Python, Spark, GeoPandas* | GitHub Oct 2024 – Dec 2024

- Developed a Python pipeline integrating APIs (OpenRouteService) and web scraping 12,000 properties to enhance real estate datasets with geospatial and amenity proximity metrics.
- Processed and cleaned large datasets using Spark; engineered livability and affordability indexes.
- Built predictive models for housing prices and visualized rental trends to support data-driven investment decisions.

New York Taxi Analysis & Prediction | *Python, Spark* | GitHub Jul 2024 – Oct 2024

- Analyzed NYC taxi data using Spark, conducting processing, feature engineering, & geospatial/temporal analysis.
- Developed predictive models (Lasso, Random Forest, Gradient-Boosted) to estimate travel cost, achieving an R-squared of 94% and an average error of only $3.80.
- Provided travel recommendations to optimize 65% cost and time based on predictive insights.

Skills

Programming Languages: Python, C++, SQL, JavaScript, Java, HTML/CSS
Frameworks & Libraries: Django, React.js, Node.js, Flask, NumPy, Pandas, Plotly, Matplotlib, PyTorch
Tools & Platforms: Docker, GCP/AWS, Git, CI/CD Pipelines, MySQL, PostgreSQL, Apache Spark, Jupyter Notebook, Google Colab, Kubernetes, Power BI